

ARDUINO CAPSTONE: DESIGN YOUR OWN IOT GADGET





HOW TO BUILD A GADGET TO STOP THIS ?

YES! AN ULTRASONIC SENSOR CHECKS THE WATER LEVEL → ARDUINO UNO DECIDES IT'S TOO FULL → BUZZER RINGS SO YOU CAN SWITCH OFF THE PUMP! 🔔

WHAT MAKES AN IOT GADGET?

- 👁️ Eyes = Sensors (Ultrasonic, LDR, PIR) — they sense the world!
- 🧠 Brain = Arduino UNO — it thinks and decides what to do.
- 👐 Hands = Actuators (Buzzers, LEDs, LCD) — they take action!

YOUR MISSION: BLUEPRINT TIME!

Choose your smart gadget project! Fill your 1-page blueprint.

Track A: Smart Jal-Rakshak (Water Overflow Alarm)

Track B: Auto-Plant Watering (Mitti Moisture Sensor)

Track C: Smart Colony Security Light (Motion Sensor)



IF WATER IS TOO HIGH,
BUZZER BEEPS! 🔔

- 🔍 A SENSOR CHECKS WHAT'S HAPPENING AROUND IT.
- 🧠 THE ARDUINO UNO BRAIN DECIDES WHAT TO DO.
- 🔔 AN ACTUATOR (BUZZER OR LIGHT) TAKES ACTION!

QUICK QUIZ: SENSORS & ACTUATORS!

1. Which device is a sensor?

A) Ultrasonic Sensor B) Buzzer C) LED

2. What does an actuator do?

A) Reads data B) Makes sound or light C) Stores code

3. If LDR senses darkness, what could happen?

A) Turn on light B) Turn off fan C) Nothing



YOUR TURN! MISTAKES = ADVENTURES!

HOW AN IOT GADGET WORKS:



A SENSOR SPOTS THE PROBLEM



ARDUINO UNO DECIDES WHAT TO DO



A BUZZER OR LIGHT TAKES ACTION!



GREAT JOB TODAY!
PRACTICE TINKERCAD & KEEP CREATING
YOUR SMART GADGETS!